arch

Wasserstein Distributionally Robust Optimization - Mathematical Formulation

The Problem

We want a model that performs well on future climate data, but we only have historical observations.

Let:

- Z = weather features (temperature, pressure, humidity, wind)
- Y = heatwave label (1 if heatwave, 0 otherwise)
- $l(\theta, Z)$ = loss function (binary cross-entropy)

Standard Empirical Risk Minimization (ERM):

$$\min_{\theta} \mathbb{E}_{Z\sim\hat{P}_n}[l(\theta, Z)]$$

where $\hat{P}_n$ is the empirical distribution of historical data.

The Issue

$\hat{P}_n$ has zero density except at observed points. If future climate shifts, the model cannot perform well.

W-DRO Formulation

Instead of trusting $\hat{P}_n$ alone, we define a ball of distributions around it and minimize the worst-case expected loss:

$$\min_{\theta} \sup_{\mathbb{Q}:W_1(\hat{P}_n,\mathbb{Q})\leq\epsilon} \mathbb{E}_{Z\sim\mathbb{Q}}[l(\theta, Z)]$$

where W_1 is the 1-Wasserstein distance and ϵ is the Wasserstein radius.

In our context: Instead of trusting only the past, we consider all possible future distributions close to the past and train against the worst one.

Wasserstein Distance

For two distributions P and $\mathbb{Q}$:

$$W_1(P, \mathbb{Q}) = \inf_{\pi \in \Pi(P, \mathbb{Q})} \int_{\mathcal{Z} \times \mathcal{Z}} \|z - \tilde{z}\| \, d\pi(z, \tilde{z})$$

This is the minimum cost of transporting probability mass from P to $\mathbb{Q}$.

Strong Duality

The infinite-dimensional inner maximization has a finite-dimensional dual:

$$\sup_{\mathbb{Q}: W_1(\hat{P}_n, \mathbb{Q}) \leq \epsilon} \mathbb{E}_{\mathbb{Q}}[l(\theta, Z)] =$$

$$\inf_{\lambda \geq 0} \left\{ \lambda \epsilon + \frac{1}{n} \sum_{i=1}^{n} \sup_{Z \in \mathcal{Z}} [l(\theta, Z) - \lambda \|Z - Z_i\|] \right\}$$

where λ is the Lagrange multiplier for the Wasserstein constraint.

The Saddle-Point Problem

$$\min_{\theta} \max_{\lambda \geq 0} \left\{ \lambda \epsilon + \frac{1}{n} \sum_{i=1}^{n} \max_{Z} [l(\theta, Z) - \lambda \|Z - Z_i\|] \right\}$$

Three components:

- Outer min (θ): model weights
- Inner max (λ): dual variable (penalty per unit of transport)
- Innermost max (Z): adversarial input

Numerical Approximation

For neural networks, the innermost $\max_Z$ is solved iteratively using Projected Gradient Descent (PGD):

$$Z^{(t+1)} = Z^{(t)} + \alpha \cdot \text{sign}\left(\nabla_Z \left[l(\theta, Z^{(t)}) - \lambda \|Z^{(t)} - Z_i\|\right]\right)$$

The Guarantee

If the true future distribution $\mathbb{Q}_{future}$ satisfies $W_1(\hat{P}_n, \mathbb{Q}_{future}) \leq \epsilon$, then:

$$\mathbb{E}_{\mathbb{Q}_{future}}[l(\theta^*_{DRO}, Z)] \leq \mathbb{E}_{\hat{P}_n}[l(\theta^*_{DRO}, Z)] + \epsilon \cdot \text{Lip}(l)$$

where $\text{Lip}(l)$ is the Lipschitz constant of the loss.

Future performance is bounded by historical performance plus the physical shift multiplied by model sensitivity.